Supplementary Materials: Cross-Architecture KV Cache Quantization

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1 Reproduction Commands

All results in the paper can be reproduced from a clean checkout of the `fajarkraton/fajarquant` repository. GPU access (NVIDIA with ≥16 GB VRAM) and a HuggingFace account with model access are required.

Listing 1: Full reproduction pipeline

```
# Clone and set up environment
git clone https://github.com/fajarkraton/fajarquant
cd fajarquant
python3 -m venv .venv
source .venv/bin/activate
pip install torch torchvision --index-url \
  https://download.pytorch.org/whl/cu124
pip install transformers datasets numpy

# Step 1: Calibrate PCA rotations (per-layer + per-head)
# Gemma (MQA, 1 head -- per-layer = per-head)
python3 scripts/calibrate_fq_v2.py \
  --model google/gemma-4-E2B \
  --output data/calibration/fq_v2_gemma_4_e2b.npz

# Mistral (GQA, 8 heads -- per-head calibration)
python3 scripts/calibrate_fq_v2.py \
  --model mistralai/Mistral-7B-v0.1 --per-head \
  --output data/calibration/fq_v2_mistral_7b_v0.1_perhead.npz

# Qwen2 (GQA, 4 heads -- per-head calibration)
python3 scripts/calibrate_fq_v2.py \
  --model Qwen/Qwen2-7B --per-head \
  --output data/calibration/fq_v2_qwen2_7b_perhead.npz

# Step 2: Profile KV heads
for model in google/gemma-4-E2B mistralai/Mistral-7B-v0.1 \
             Qwen/Qwen2-7B; do
  python3 scripts/profile_kv_heads.py --model "$model"
done

# Step 3: Tune thresholds + generate strategies
python3 scripts/tune_thresholds.py \
  --model mistralai/Mistral-7B-v0.1 --bits 3,4 \
  --calibration data/calibration/fq_v2_mistral_7b_v0.1_perhead.npz
python3 scripts/tune_thresholds.py \
  --model Qwen/Qwen2-7B --bits 3,4 \
  --calibration data/calibration/fq_v2_qwen2_7b_perhead.npz

# Step 4: PPL-guided 2-bit strategy selection
for model in google/gemma-4-E2B mistralai/Mistral-7B-v0.1 \
             Qwen/Qwen2-7B; do
  python3 scripts/ppl_guided_select.py --model "$model"
done

# Step 5: Full perplexity evaluation (Table 7)
```

```
# Gemma (seq_len=2048)
for bits in 2 3 4; do
  python3 scripts/eval_perplexity_v3.py \
    --model google/gemma-4-E2B \
    --strategy data/calibration/strategy_gemma_4_e2b_${bits}bit*.json \
    --calibration data/calibration/fq_v2_gemma_4_e2b.npz \
    --bits $bits --seq-len 2048 --max-samples 30 \
    --output data/kv_cache/perplexity_v3.1_gemma_${bits}bit.json
done

# Mistral (seq_len=2048)
for bits in 2 3 4; do
  python3 scripts/eval_perplexity_v3.py \
    --model mistralai/Mistral-7B-v0.1 \
    --strategy data/calibration/strategy_mistral_7b_v0.1_${bits}bit*.json \
    --calibration data/calibration/fq_v2_mistral_7b_v0.1_perhead.npz \
    --bits $bits --seq-len 2048 --max-samples 30 \
    --output data/kv_cache/perplexity_v3.1_mistral_${bits}bit.json
done

# Qwen2 (seq_len=1024 due to OOM at 2048)
for bits in 2 3 4; do
  python3 scripts/eval_perplexity_v3.py \
    --model Qwen/Qwen2-7B \
    --strategy data/calibration/strategy_qwen2_7b_${bits}bit*.json \
    --calibration data/calibration/fq_v2_qwen2_7b_perhead.npz \
    --bits $bits --seq-len 1024 --max-samples 30 \
    --output data/kv_cache/perplexity_v3.1_qwen2_${bits}bit.json
done

# Step 6: Verify paper tables
python3 scripts/verify_paper_tables.py --strict
```

Expected runtime: Steps 1–4 require ∼4 GPU hours (dominated by calibration). Step 5 requires ∼4 GPU hours (∼30 minutes per model×bit evaluation). Total: ∼8 GPU hours on RTX 4090.

2 Dataset Provenance

2.1 Evaluation Dataset

Table 1: Evaluation dataset

Dataset	Salesforce/wikitext (wikitext-2-raw-v1)
Split	test
Tokens (Gemma/Mistral, L=2048)	61,410 (30 chunks)
Tokens (Qwen2, L=1024)	30,690 (30 chunks)
License	CC BY-SA 3.0
Source	`https://huggingface.co/datasets/Salesforce/wikitext`

2.2 Calibration Data

PCA rotation matrices and per-head statistics are calibrated on the first 128 samples from WikiText-2 training split, processed through each model's attention layers. Calibration is performed once and reused for all evaluations.

3 Model Provenance

All models are publicly available on HuggingFace. Gemma 4 E2B requires accepting Google's Gemma license agreement. Mistral 7B and Qwen2-7B are Apache 2.0 licensed with no access restrictions.

Table 2: Calibration data checksums (SHA-256, first 16 hex chars)

File	Size	SHA-256 prefix
`fq_v2_gemma_4_e2b.npz`	8.2 MB	`b4d43463daad65c0`
`fq_v2_mistral_7b_v0.1.npz`	2.8 MB	`ad791e983d539689`
`fq_v2_mistral_perhead.npz`	23 MB	`c4518c63a0ca7f41`
`fq_v2_qwen2_7b.npz`	2.5 MB	`834f206fcd665425`
`fq_v2_qwen2_perhead.npz`	9.8 MB	`d7c5e9daa83a7b2c`

Table 3: Models used in evaluation

Model	HuggingFace ID	License	KV Heads
Gemma 4 E2B	`google/gemma-4-E2B`	Gemma	1 (MQA)
Mistral 7B v0.1	`mistralai/Mistral-7B-v0.1`	Apache 2.0	8 (GQA)
Qwen2 7B	`Qwen/Qwen2-7B`	Apache 2.0	4 (GQA)

4 Hardware and Software Environment

Table 4: Evaluation environment

CPU	Intel Core i9-14900HX (24 cores / 32 threads)
GPU	NVIDIA GeForce RTX 4090 Laptop (16,376 MiB)
RAM	32 GB DDR5
OS	Ubuntu 24.04, Linux 6.17.0-20-generic
GPU Driver	590.48.01
Python	3.12
PyTorch	2.6.0+cu124
Transformers	5.5.3
NumPy	2.4.3
Datasets	4.8.4
Rust	1.87 (for native Fajar Lang benchmarks)

5 Qwen2 Sequence Length Note

Qwen2-7B evaluations use L=1024 instead of L=2048 due to GPU memory constraints. At L=2048, the model triggers CUDA OOM on our 16 GB RTX 4090 Laptop GPU because Qwen2's 28-layer, 28-head architecture requires more activation memory than Gemma (15 layers) or Mistral (32 layers but efficient GQA-8). The shorter sequence length means Qwen2 perplexity values are not directly comparable in absolute terms to Gemma/Mistral, but the *relative ordering* between methods (FQ v3.1 vs KIVI vs TQ) is valid within the Qwen2 column.

6 Code and Data Availability

- **Algorithm + scripts:** `https://github.com/fajarkraton/fajarquant`
- **Native implementation:** `https://github.com/fajarkraton/fajar-lang` (`src/runtime/ml/`)
- **Kernel port:** `https://github.com/fajarkraton/fajaros-x86` (`kernel/compute/fajarquant.fj`)
- **Calibration data:** included in the `fajarquant` repo under `data/calibration/`
- **Evaluation results:** included under `data/kv_cache/perplexity_v3.1_*.json`